

SULTANATE OF OMAN

TARBIYAT

NATIONAL YOUTH TECH MISSIONS SCHOOL & COLLEGE

NIZWA, AL DAKHILIYAH GOVERNORATE, SULTANATE OF OMAN

Version 5.0 - The Comprehensive Sovereign Implementation Framework

"From India with Respect - For Oman with Commitment"

Concept Originated by: Dr. Sudhir Pawar (alias Dr. Salahuddin)

20+ Years in Educational Institutional Development | INSPIRE Method | Institutional Compliance Manual & Handbook for India

Submitted To	Ministry of Education MoHERI MOCIIP - Sultanate of Oman
Submitted By	Tarbiyat Sociotech Foundation, New Delhi, Republic of India
Version	5.0 - Comprehensive Sovereign Implementation Framework
Investment	OMR 39,250,000 (approx. Rs 857 Crore) - Fully Government of Oman Funded
Boarding	100% Full Residential - Day-Boarding Prohibited
Ownership	100% Government of Oman - All Assets, Land, Data, IP
Engagement	Minimum 10 Years - Reviewed Mutually
Launch	September 2028 - Nizwa
Annexures	A: HR B: Financial Ops C: Security D: Engineering

Tarbiyat Sociotech Foundation, New Delhi, Republic of India

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EXECUTIVE SUMMARY - VERSION 5.0



THE DEFINITIVE BOARDROOM IMPLEMENTATION PACKAGE

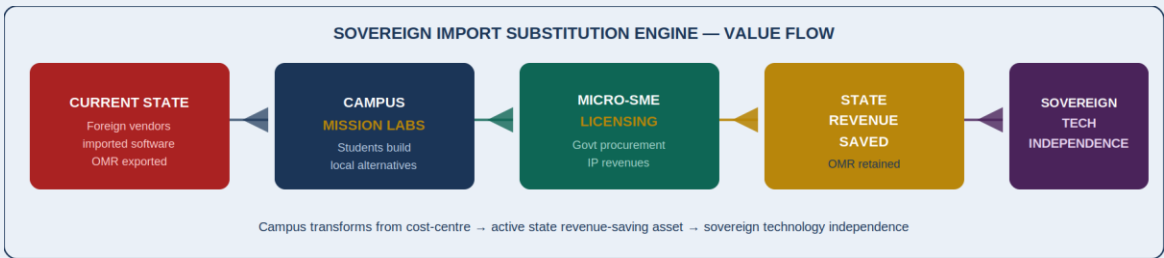
Tarbiyat National Youth Tech Missions School & College is Oman's first Government-Owned National Youth R&D Campus. Version 5.0 - The Comprehensive Sovereign Implementation Framework - is the definitive, Ministry-ready implementation document. It preserves all seven strategic breakthroughs of Version 4.0: (1) Sovereign Competency Graph replacing all examinations; (2) Student Micro-Venture Economy under Omani law; (3) Industry-in-Residence daily co-working; (4) Siyadah Al-Raqmiyya cyber-sovereignty framework; (5) Sovereign Import Substitution Engine; (6) Parallel Dual-Credentialing with board exam safety net; (7) 100% Full Boarding with Separation of Concerns dual faculty architecture. Four Technical Annexures extend the document with institutional-grade implementation blueprints for formal government evaluation.

Project	Tarbiyat National Youth Tech Missions School and College, Nizwa
Concept Author	Dr. Sudhir Pawar (Dr. Salahuddin) - INSPIRE Method, 20+ years institutional development
Campus	1,500 School + 400 College + 500-Bed Full Boarding
Site	60,000 sqm 30,000 sqm built-up G+4 LEED Certified
Investment	OMR 39,250,000 (~Rs 857 Crore) Engineering Blueprint: OMR 60,250,350
Ownership	100% Government of Oman from Day 1
Boarding	100% Full Residential - Day-Boarding Strictly Prohibited
Credentialing	Proof of Capability Profile + National Board Examinations
Launch	September 2028
Annexure A	Premium Human Resource and Faculty Architecture
Annexure B	Financial Operations and Recurring Overhead Tracker
Annexure C	Holistic Physical, Safety and Data Security Playbook
Annexure D	Engineering, Infrastructure and Lab Equipment Master Plan

STRATEGIC PROPOSAL NARRATIVE (Sections 1-16)

Full Strategic Narrative - Preserved from Version 4.0

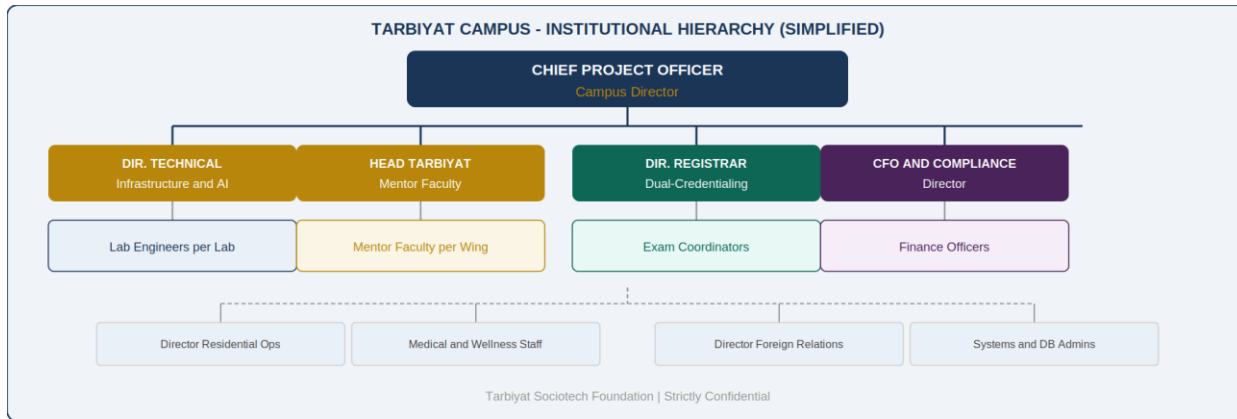
The complete strategic narrative from Version 4.0 - covering Vision and Mission, Oman Vision 2040 alignment, seven structural breakthroughs, four-phase academic model with dual-credentialing, college and micro-venture economy, Tech and Tarbiyat four-circle framework, world-class campus design, governance and Omanization, implementation team and dual-faculty separation of concerns, collaborations, 24-month roadmap, financial summary, expected outcomes, risk assessment, and nine formal requests - is presented in the master Version 4.0 document, which forms Part I of this complete proposal package. Version 5.0 constitutes Part II: the Technical Annexures package, extending the strategic narrative with full institutional-grade implementation detail.



Annexure A	HR and Faculty Architecture: Full institutional hierarchy, 15 post profiles, staff qualifications, and headcount summary (170 permanent staff)
Annexure B	Financial Operations Tracker: 3-year and 5-year technology amortization cycles, premium facility management, catering logistics, HVAC, and full recurring OPEX consolidated at OMR 2.9M per year from Year 3
Annexure C	Security Playbook: Four-layer physical and cyber-sovereignty architecture, visitor sandboxing, biometric access matrix, and sovereign data isolation protocols for all corporate and government datasets
Annexure D	Engineering Master Plan: Complete spatial allocation, equipment lists, and cost estimates across all 60,000 sqm - 12 labs, 24 classrooms, auditorium, residential, sports, and all sovereign utility networks

TECHNICAL ANNEXURE A PREMIUM HUMAN RESOURCE AND FACULTY ARCHITECTURE

A

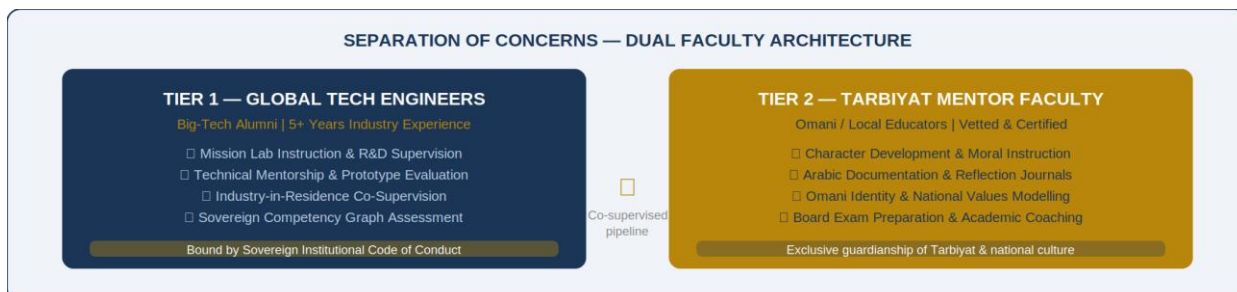


A.1 - Introduction and Institutional Mandate

The Tarbiyat National Youth Tech Missions Campus operates as a 24-hour, 365-day residential institution serving 1,900 students across school and college programmes, supported by 170-plus permanent resident staff. The human resource architecture of this campus is not designed for a conventional school. It is designed for a sovereign R&D institution that simultaneously functions as a world-class boarding school, a national youth innovation laboratory, an IP commercialisation centre, and a character development institution grounded in Omani Islamic values.

The institutional hierarchy is structured in four formal tiers: Executive Leadership, Operational Directors, Campus Life and Wellbeing, and Engineering and Technical Support. All roles operate under the Sovereign Institutional Code of Conduct - a legally binding document governing the conduct, mandate boundaries, and confidentiality obligations of every member of staff - particularly as they relate to the Separation of Concerns between Tier 1 Global Tech Engineers and Tier 2 Tarbiyat Mentor Faculty.

A.2 - Tier 1: Executive Leadership Block



Post A.2.1 - Chief Project Officer / Campus Director

Chief Project Officer / Campus Director

Grade: Executive Grade 1 **Reports to:** Governing Board of the Tarbiyat National Youth Tech Missions Campus

Qualifications Required: Minimum 20 years senior leadership in educational institution management, national technology policy, or government-linked enterprise. MBA or equivalent postgraduate qualification in leadership, public administration, or institutional governance. Prior experience managing residential campuses or R&D institutions strongly preferred.

Scope: Full institutional accountability for all academic, operational, financial, security, and reputational outcomes. Primary liaison with Governing Board, Ministry of Education, MoHERI, and MOCIIP. Final authority on all HR decisions, capital expenditures, partnership agreements, and policy implementations.

- Chairs the weekly Campus Leadership Council and monthly Governing Board reporting session.
- Signs all institutional MOUs, Micro-SME legal agreements, Industry-in-Residence Protocols, and Sovereign Institutional Codes of Conduct.
- Accountable for Omanization targets, import substitution metrics, dual-credentialing outcomes, and annual Proof of Capability Profile issuance.
- Represents the campus at Ministry-level meetings, international technology policy forums, and national innovation showcases.
- Oversees the 10-year implementation partnership agreement with Tarbiyat Sociotech Foundation and manages Foundation-Government boundary compliance.

Post A.2.2 - Director of Technical Infrastructure and AI Systems

Director of Technical Infrastructure and AI Systems

Grade: Executive Grade 2 **Reports to:** Chief Project Officer / Campus Director

Qualifications Required: Minimum 15 years in enterprise technology leadership, ideally at a big-tech company (Google, Microsoft, IBM, Siemens, or equivalent). Deep expertise in AI systems architecture, cloud and on-premise data infrastructure, cybersecurity frameworks, and sovereign data systems. Postgraduate qualification in Computer Science, Data Engineering, or Cybersecurity. Experience designing Tier-2 data centres preferred.

Scope: Full operational responsibility for the campus Tier-2 Data Centre, Sovereign Competency Graph AI engine, Proof of Capability Profile platform, all 12 mission lab technology systems, campus-wide Wi-Fi 6 and fibre infrastructure, smart campus management systems, and Siyadah Al-Raqmiyya cybersecurity curriculum technical content.

- Designs, commissions, and maintains the Sovereign Competency Graph AI engine - the most critical single technology system on campus.
- Ensures all student data, mission IP, and corporate R&D datasets are housed exclusively within Omani sovereign jurisdiction on the on-campus Tier-2 Data Centre.
- Leads the technology amortization planning cycle - managing 3-year and 5-year hardware refresh schedules for all lab equipment and network systems.
- Co-supervises the Industry-in-Residence Hub - ensuring corporate partner data access is firewalled and sandboxed from campus sovereign data systems.
- Oversees the Cybersecurity Operations Centre lab and serves as technical advisor for all Siyadah Al-Raqmiyya student missions.

Post A.2.3 - Head of Omani Tarbiyat Mentor Faculty

Head of Omani Tarbiyat Mentor Faculty

Grade: Executive Grade 2 **Reports to:** Chief Project Officer / Campus Director

Qualifications Required: Omani national or Arab national with deep roots in Omani Islamic and cultural tradition. Minimum 15 years experience in Islamic education, character development, or values-based institutional leadership. Formal qualifications in Islamic Studies, Arabic Language, or Educational Leadership. Prior experience managing residential boarding school environments. Fluent in Arabic and English. Deep familiarity with Omani national curriculum and Ministry of Education academic frameworks.

Scope: Exclusive institutional authority over all character development, national identity formation, Arabic language programming, Islamic studies, and Tarbiyat residential programming on campus. Responsible for selection, training, supervision, and performance management of all Tier 2 Tarbiyat Mentor Faculty. Ensures Separation of Concerns principle is maintained at all times.

- Designs and delivers the Tarbiyat Evening Programme - the structured residential programme operating daily after mission hours.
- Selects and vets all Tarbiyat Mentor Faculty to the highest standards of Islamic knowledge, personal character, and youth development competency.
- Co-supervises all student mission debriefs - evaluating the Tarbiyat and Ethics Quotient component alongside technical evaluation by Tier 1 engineers.
- Leads Friday Juma coordination, Ramadan programming, community service calendars, and Omani heritage cultural events.
- Reports violations of the Sovereign Institutional Code of Conduct by Tier 1 staff directly to the Chief Project Officer.
- Oversees the Arabic reflection journal programme - ensuring every student maintains substantive Tarbiyat documentation feeding into the Sovereign Competency Graph.

A.3 - Tier 2: Operational Directors Layer

Post A.3.1 - Director of Registrar and Dual-Credentialing Examination Mapping

Director of Registrar and Dual-Credentialing Examination Mapping

Grade: Senior Director Grade A **Reports to:** Chief Project Officer / Campus Director

Qualifications Required: Minimum 12 years in academic administration, curriculum alignment, or examination board management. Prior experience with CBSE, Cambridge International, or equivalent national examination frameworks. Postgraduate qualification in Educational Management or Curriculum Design. Familiarity with MoHERI accreditation and Omani academic recognition frameworks.

Scope: Institutional management of all student academic records, the dual-credentialing system, board examination registrations and logistics, MoHERI Proof of Capability Profile submissions, and the ongoing reverse-engineering of board syllabi into daily skills sprint frameworks.

- Maintains the master syllabus mapping matrix - ensuring every mission topic in Phases 1 through 4 is cross-referenced with its corresponding board examination learning objective.
- Manages all student registrations with external examination boards (CBSE, Cambridge, or Omani national board) for Grade 10 and Grade 12 examinations.
- Coordinates the external examination logistics: invigilation, secure answer script handling, result processing, and certification issuance.
- Submits Proof of Capability Profile dossiers to MoHERI annually for formal recognition update.
- Generates the annual dual-credentialing outcomes report for the Governing Board and Ministry of Education.

Post A.3.2 - Chief Financial Officer and Compliance Director

Chief Financial Officer and Compliance Director

Grade: Senior Director Grade A **Reports to:** Chief Project Officer / Campus Director

Qualifications Required: Chartered Accountant or equivalent postgraduate qualification in Finance, Public Accounting, or Institutional Financial Management. Minimum 12 years in senior financial roles, preferably within government-funded educational or infrastructure institutions. Familiarity with Omani financial regulations, government procurement frameworks, and MOCIIP reporting requirements.

Scope: Full financial accountability for the campus including CAPEX drawdown management, OPEX tracking, payroll, procurement compliance, IP revenue distribution, import substitution savings reporting, and annual audit preparation. Manages the Foundation-Government financial boundary.

- Manages the milestone-based CAPEX disbursement schedule - ensuring construction and commissioning funds are drawn down against verified project milestones.
- Oversees the IP revenue distribution framework: 40% to student teams, 40% to MoHERI national education innovation fund, 20% to campus operational reinvestment.
- Prepares the annual import substitution fiscal savings report for MOCIIP - quantifying OMR value of foreign technology procurement replaced by student Micro-SME solutions.
- Manages the 3-year and 5-year technology amortization and hardware renewal budgets in coordination with the Director of Technical Infrastructure.
- Ensures full compliance with Omani public audit requirements, Foundation-Government contractual financial terms, and Governing Board financial reporting obligations.

Post A.3.3 - Director of Foreign Relations and Tech Policy Delegations

Director of Foreign Relations and Tech Policy Delegations

Grade: Senior Director Grade B **Reports to:** Chief Project Officer / Campus Director

Qualifications Required: Minimum 10 years in international relations, technology diplomacy, or government liaison roles. Prior experience representing organisations at ITU, GovTech, UNESCO, or similar international forums strongly preferred. Postgraduate qualification in International Relations, Public Policy, or Technology Governance. Fluency in English and Arabic required.

Scope: Management of all international partnerships, student delegation preparation for technology policy forums, Industry-in-Residence Partner Relations, Siyadah Al-Raqmiyya international programme coordination, and bilateral institutional agreements with global academic and technology organisations.

- Identifies and secures international representation opportunities for Grade 11 and 12 students at ITU, GovTech conferences, AI governance dialogues, and regional technology forums.
- Manages Industry-in-Residence Partner Agreements - coordinating rotating engineer commitments from PDO, OQ, Omantel, Asyad, and Royal Oman Police.
- Coordinates with the National Cybersecurity Authority and international AI governance bodies on Siyadah Al-Raqmiyya curriculum validation.
- Manages all international academic partnerships including UTAS Nizwa, Indian institutions, Siemens, Microsoft, Google, IBM, and Cisco.
- Represents the campus at bilateral government meetings as the formal institutional liaison for foreign partnership matters.

A.4 - Tier 3: Hospitality and Campus Life Block

Post A.4.1 - Director of Residential Operations and Campus Wellbeing**Director of Residential Operations and Campus Wellbeing****Grade:** Director Grade B **Reports to:** Chief Project Officer / Campus Director**Qualifications Required:** Minimum 10 years managing large residential institutions - boarding schools, university halls of residence, or equivalent 24-hour residential facilities. Experience managing gender-segregated residential environments. Postgraduate qualification in Hospitality Management, Residential Education, or Institutional Operations. Strong knowledge of adolescent welfare frameworks and Omani cultural expectations.**Scope:** Complete operational responsibility for the 500-bed student hostel, 50-plus faculty and staff residences, all dining facilities, student health centre, sports and recreation complex, and the Tarbiyat Evening Programme logistics.

- Manages the daily boarding schedule - ensuring the triple-stream academic architecture is operationally supported by the residential timetable.
- Oversees gender-segregated hostel wing management - ensuring strict biometric access control and complete physical separation of male and female residential zones at all times.
- Manages all campus dining operations including Halal bulk catering logistics, 800-seat concurrent dining session scheduling, dietary monitoring, and vendor compliance.
- Coordinates the campus wellbeing programme with Resident Medical Staff and Adolescent Development Mentors.
- Manages the sports and recreation complex timetable - ensuring mandatory daily physical activity for all residential students.

Posts A.4.2 - Resident Medical and Wellness Staff

The campus operates a 24-hour on-site medical facility staffed by the following permanently resident medical professionals:

Post	Qualification	Shift	Scope
Senior Resident Medical Officer	MBBS or MD, minimum 5 years clinical experience	24x7 on-call (rotating)	Primary care, emergency response, referral coordination, adolescent health oversight
Resident General Practitioner	MBBS, minimum 3 years	Day shift daily	Routine consultations, sick bay management, health screenings, vaccination records
Registered Nurse (x3)	Diploma or Degree Nursing	Three rotating 8-hour shifts	Routine nursing care, medication administration, first aid, sick bay monitoring
Adolescent Mental Health Mentor (x2)	BSc or MSc Psychology or Counselling	Day and evening availability	Individual and group counselling, stress support, academic pressure management, crisis protocol
Physiotherapist	Diploma or Degree Physiotherapy	5 days per week day shift	Sports injury management, rehabilitation, posture and wellness programmes
Pharmacist	BPharm minimum	6 days per week day shift	Dispensary management, prescription fulfilment, controlled substance compliance

A.5 - Tier 4: Engineering and Technical Support Staff

Post	Count	Qualifications Required	Primary Responsibilities
Senior Systems Administrator	2	BSc Computer Science, 7+ years enterprise sysadmin, Cisco CCNA/CCNP preferred	Full campus network administration: fibre backbone, Wi-Fi 6, enterprise switches, VLANs, firewall rule management, and Data Centre operating environment
Sovereign Data Centre Manager	1	BTech or MSc Computer Science, 10+ years data centre operations, certified in DCIM platforms	24x7 uptime management of Tier-2 Data Centre: server racks, cooling, UPS, backup systems, and Sovereign Competency Graph database continuity
Database Analyst and Graph Engineer	2	MSc Data Science or Database Administration, experience with graph databases and AI/ML pipeline management	Operates and maintains the Sovereign Competency Graph AI engine: data ingestion from mission logs, competency scoring, anomaly detection, and Proof of Capability Profile generation
Power Grid Supervisor	2	BTech Electrical Engineering, 8+ years power systems, familiarity with solar-battery-grid hybrid systems	24x7 oversight of 500 kW solar array, 1 MWh battery storage, diesel generator backup, and campus power distribution: load balancing, fault response, and grid health monitoring
Solar and Energy Systems Technician	3	Diploma or BTech Electrical, certified solar PV technician	Routine maintenance of solar panels, inverters, battery management systems, and grid-tie connections; performance data logging and efficiency reporting
Advanced Lab Technician (per lab)	12	Diploma or BTech in relevant technology domain per lab assignment	Daily setup, maintenance, calibration, and safety management of all equipment in each of the 12 mission labs; student safety induction; equipment fault escalation
HVAC and Chiller Systems Technician	2	Diploma Mechanical or HVAC Engineering, 5+ years with large commercial chiller systems	Maintenance and operation of dual central AC chiller systems and cooling towers serving 30,000 sqm built campus; preventive maintenance; refrigerant compliance
Water Treatment Plant Operator	1	Diploma in Water Treatment or Civil Engineering, 5+ years experience	Daily operation of campus water desalination and treatment plant: water quality testing, chemical dosing, storage tank management, and distribution system compliance
Smart Campus Systems Engineer	1	BTech Electrical or IoT Engineering, 5+ years with BMS/SCADA platforms	Operates and maintains the AI-driven Building Management System: energy monitoring, access

			control, CCTV network, environmental sensors, and automated maintenance scheduling
Security Command Centre Supervisor	2 (rotating)	Military or police background preferred, 8+ years institutional security management	Manages four-tower perimeter security command network: CCTV analytics oversight, visitor sandboxing workflow, biometric access log review, incident escalation, and daily security briefings

A.6 - Total Permanent Headcount Summary

Staff Category	Headcount	Residency Status
Executive Leadership Block (CPO, Tech Director, Head Tarbiyat Faculty)	3	Fully Resident on Campus
Operational Directors (Registrar, CFO, Foreign Relations)	3	Fully Resident or Near-Campus
Academic Mission Coaches and Lab Instructors (Tier 1 Global Tech Engineers)	24	Rotating - minimum 2 years per placement
Tarbiyat Mentor Faculty (Tier 2 - per hostel wing and subject area)	20	Fully Resident on Campus
Registrar and Examination Office	6	On-Campus Daily
Finance, Procurement and Compliance Office	5	On-Campus Daily
Foreign Relations and Partnership Office	3	On-Campus and Travel-Based
Director Residential Ops and Hostel Management Team	8	Fully Resident on Campus
Medical and Wellness Staff (Doctors, Nurses, Psychologists, Physio)	10	Rotating 24-Hour Residential Coverage
Engineering and Technical Support Staff (Sysadmin, Data, Power, Labs, HVAC)	30	On-Campus Daily with On-Call Roster
Security Command Centre and Perimeter Team	12	24x7 Rotating Shift - On Campus
Dining, Catering and Housekeeping	24	On-Campus Daily Shifts
Grounds, Landscaping and Maintenance	12	On-Campus Daily
Administrative and Student Services Support	10	On-Campus Daily
TOTAL PERMANENT STAFF	170	Resident or Daily On-Campus

TECHNICAL ANNEXURE B
FINANCIAL OPERATIONS AND RECURRING OVERHEAD TRACKER

B

B.1 - Introduction and Financial Sustainability Framework

The financial architecture of the Tarbiyat National Youth Tech Missions Campus extends well beyond the initial Capital Expenditure phase. As a 24-hour, 365-day residential R&D institution operating world-class technology infrastructure, the campus carries significant recurring operational cost obligations across three categories: technology amortization and lifecycle renewal; premium facility management and green utilities upkeep; and human resource operational costs. This Annexure provides a detailed, realistic financial breakdown of ongoing operational expenditure beyond the initial CAPEX setup, enabling the Government of Oman to plan multi-year budget allocations with institutional-grade precision.

B.2 - Technology Amortization and Renewal Lifecycle

B.2.1 - Three-Year Hardware Replacement Cycle

Technology assets in mission-critical, high-utilisation research and laboratory environments have significantly shorter useful lives than standard commercial IT infrastructure. The following categories require structured three-year replacement or major refurbishment cycles, budgeted from Year 3 of operation onward:

Technology Asset Category	Unit Count	Unit Cost (OMR)	3-Year Total (OMR)	Rationale
GPU Computing Workstations (AI and ML Labs)	24	1,800	43,200	GPU generations advance every 18-36 months; workstations become bottlenecks for advanced AI mission work after 3 years of intensive use
High-Performance Computing Server Nodes (Data Centre)	8	9,500	76,000	Server nodes in continuous operation require replacement at 3-year intervals to maintain Sovereign Competency Graph processing speeds
Industrial Robotics Controllers and End-of-Arm Tooling	12 sets	3,200	38,400	Mechanical wear in 6-axis robotic arms and CNC controller firmware requires tooling replacement and controller software upgrade on 3-year schedule
Network Core and Distribution Switch Gear	18 units	2,100	37,800	Enterprise managed switches require replacement to maintain 10 Gbps

				throughput and Wi-Fi 6 compatibility as standards evolve
3D Printing Systems (FDM, SLA, SLS Printers)	16	1,400	22,400	Print heads, build platforms, and laser units degrade under continuous student use; replacement maintains dimensional accuracy for mission prototypes
Drone Fleet and UAV Systems	24 units	900	21,600	Multi-rotor mission drones have average useful lives of 250-400 flight hours under student training conditions; battery cells degrade significantly within 3 years
AR/VR Headset Arrays and Motion Capture Systems	40 headsets	600	24,000	AR/VR headsets have a practical mission-lab useful life of 2-3 years before lens degradation, sensor drift, and software ecosystem obsolescence
Biotech and Medical Lab Equipment (analytical)	Per lab	8,500 lot	8,500	Centrifuges, spectrophotometers, PCR units require calibration, service, and component replacement on 3-year maintenance contracts
3-YEAR CYCLE TOTAL TECHNOLOGY RENEWAL	—	—	271,900	Annualised provision: OMR 90,633 per year - allocated to Technology Renewal Reserve from Year 1

B.2.2 - Five-Year Hardware Replacement and Major Infrastructure Renewal

Technology Asset Category	5-Year Renewal Cost (OMR)	Scope of Renewal
Sovereign Competency Graph AI Engine - Server Rack Refresh and Software Upgrade	95,000	Full rack replacement, OS migration, database engine upgrade, and AI model retraining pipeline refresh incorporating 5 years of pedagogical learning and updated national syllabus frameworks
Solar PV Panel Array Replacement (30% of panels - degradation-based)	85,000	Monocrystalline silicon panels degrade approximately 0.5% per year; 5-year partial replacement of lowest-performing panels in

		the 500 kW array maintains peak generation capacity
Battery Energy Storage System - Cell Replacement (1 MWh)	140,000	LFP battery cells degrade to approximately 70% capacity after 2,500-3,000 charge cycles (approximately 5-7 years at campus utilisation levels); cell module replacement extends system life
CNC Milling Centres and Fabrication Mechatronics (spindle and axis rebuild)	48,000	Industrial CNC machines in continuous use require spindle bearing replacement, ball screw reconditioning, and servo motor overhaul at 5-year intervals to maintain machining precision
Enterprise Wi-Fi 6 Access Point Network - Full Refresh	38,000	Campus-wide Wi-Fi 6 infrastructure (96 access points across 30,000 sqm) requires full technology generation refresh at 5-year intervals as Wi-Fi 7 becomes campus-networking baseline
Photovoltaic Test Chamber and Solar Lab Equipment	22,000	Solar materials characterisation equipment (I-V curve tracers, EL imaging cameras, environmental chambers) requires calibration overhaul and component replacement at 5-year service intervals
Enterprise Software Licensing Pools - Five-Year Renewal	55,000	Adobe Creative Suite, Autodesk Engineering Suite, MATLAB, ArcGIS, Cybersecurity Lab platforms, AI development environments, and LMS platform - 5-year enterprise agreement renewal
Micro-Grid Testbed and Power Electronics Lab Equipment	30,000	Power electronics lab: grid simulators, power analysers, variable frequency drives, and inverter test benches require major calibration overhaul and component replacement at 5-year intervals
5-YEAR CYCLE TOTAL CAPITAL RENEWAL	513,000	Annualised provision: OMR 102,600 per year - allocated to Capital Renewal Reserve from Year 1 of operation

B.3 - Premium Facility Management and Green Utilities Upkeep

B.3.1 - 500 kW Solar Array and 1 MWh Battery System Annual Operations

Maintenance Item	Annual Cost (OMR)	Scope and Frequency
Solar Panel Professional Cleaning (quarterly)	4,800	Quarterly professional cleaning of 2,100-plus panels using deionised water and non-abrasive equipment; critical in Oman's dusty climate where soiling losses reach 1-2% per week without maintenance
Inverter and Power Electronics Annual Service	6,200	Annual professional service of string and central inverters: capacitor inspection, firmware updates,

		connection torque verification, and thermal imaging of hot spots
Solar Monitoring System Subscription and Calibration	2,400	Annual subscription for cloud-based solar performance monitoring platform; pyranometer calibration; yield loss alert system maintenance
Battery Management System Software and Annual Service	3,800	Annual BMS firmware update, cell balancing calibration, thermal management system service, and State-of-Health diagnostic reporting for the 1 MWh storage system
Diesel Generator Annual Service and Fuel Reserve	8,500	Annual service: oil change, filter replacement, injector testing, load bank test; maintenance of 15,000-litre diesel reserve for 72-hour autonomous operation
ANNUAL SOLAR AND STORAGE OPERATIONS TOTAL	25,700	This figure does not include the 5-year battery cell replacement cycle budgeted in B.2.2

B.3.2 - Premium Halal Bulk Catering Operations (800-Seat Concurrent Dining)

Catering Cost Category	Annual Estimate (OMR)	Notes
Food and Beverage Procurement (Halal-Certified Bulk Institutional)	420,000	Based on OMR 220 per student per month for 1,500 boarding students and 170 staff; includes breakfast, lunch, dinner, and Friday special meals; all suppliers must hold GCC Halal certification
Kitchen Operations and Cooking Labour	96,000	12 trained kitchen staff: Head Chef (Halal-certified institutional catering), three sous-chefs (Arabic, Asian, continental cuisines), six kitchen assistants, and two cold-chain logistics staff
Dining Hall Operations and Service Labour	48,000	Eight dining service staff per shift, three shifts daily; professional linen and crockery management; dining hall deep-clean twice weekly; special event catering setup
Kitchen Equipment Maintenance and Replacement	18,000	Annual preventive maintenance contract for commercial cooking equipment: combi-ovens, blast chillers, industrial dishwashers, cold rooms, and walk-in freezers; 10% equipment replacement provision
Food Safety, HACCP Compliance, and Audit	6,000	Annual third-party HACCP compliance audit, food safety certification renewal, water quality testing, and pest control contract
Special Dietary and Medical Nutrition Management	12,000	Provision for students and staff with medical dietary requirements, food allergies, or religious dietary specifications beyond standard Halal

ANNUAL HALAL CATERING OPERATIONS TOTAL	600,000	Per capita annual cost: approximately OMR 315 per resident per year - in line with GCC institutional boarding school benchmarks
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B.3.3 - Native Green Landscaping and Grounds Management

Item	Annual Cost (OMR)	Scope
Native Omani and Arid-Adapted Landscaping Maintenance	18,000	Monthly professional grounds maintenance: pruning of native Ghaf, Sidr, Neem, and date palm species; seasonal replanting; drip irrigation system inspection and emitter replacement
Smart Drip Irrigation System Operations	9,600	Automated soil-moisture sensor-triggered drip irrigation across all green zones; monthly sensor calibration; annual system audit; integrated with Smart Campus Management System
Greywater Recycling System Operations	6,800	Monthly servicing of campus greywater treatment and recycling plant; water quality compliance testing; pumping system maintenance; filtered greywater directed exclusively to irrigation circuits
Outdoor Sports Surface Maintenance	12,000	Annual resurfacing of outdoor sports courts; running track lane marking refresh; natural grass sports field care including aeration, fertilisation, and mowing; swimming pool chemical balance and filtration service
ANNUAL LANDSCAPING AND GROUNDS TOTAL	46,400	Includes all outdoor recreational and green infrastructure maintenance excluding sports complex building services

B.3.4 - Dual Central AC Chiller System Annual Operations

Item	Annual Cost (OMR)	Scope
Chiller Preventive Maintenance Contract (semi-annual)	28,000	Two comprehensive chiller services per year by certified HVAC engineers: refrigerant charge verification, oil analysis, condenser tube brushing, compressor bearing inspection, and controls calibration for both primary and backup chiller units
Cooling Tower Water Treatment and Chemical Programme	14,400	Monthly water treatment chemical supply and dosing for cooling tower circuits: biocide, scale inhibitor, and corrosion inhibitor programme; quarterly Legionella testing and compliance documentation
Air Handling Unit and Fan Coil Unit Service	18,000	Bi-annual service of all AHUs and FCUs across 30,000 sqm: filter replacement, coil cleaning, belt and bearing inspection across 24 smart

		classrooms, 12 labs, auditorium, and residential blocks
Building Automation System Service and Optimisation	8,400	Annual BAS software update and control optimisation: demand-controlled ventilation scheduling, chiller staging sequences, overnight setback programming, and thermal comfort monitoring integration
ANNUAL HVAC AND CHILLER OPERATIONS TOTAL	68,800	Per-sqm annual HVAC cost: approximately OMR 2.29 per sqm - consistent with GCC high-performance institutional benchmarks

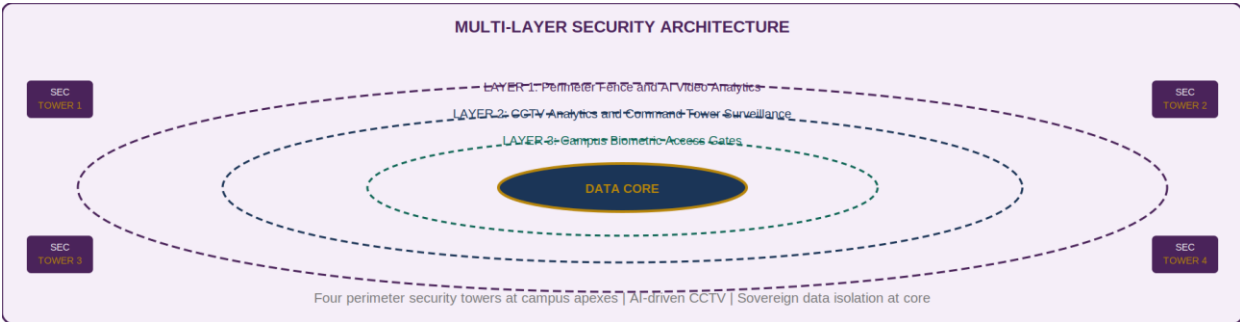
B.4 - Annual Recurring OPEX Consolidated Summary (from Year 3)

OPEX Category	Annual Cost (OMR)
Human Resource - Total Payroll (170 permanent staff, Year 3+ rates)	1,480,000
Technology Renewal Reserve (3-year cycle annualised provision)	90,633
Capital Renewal Reserve (5-year cycle annualised provision)	102,600
Solar and Battery System Annual Operations	25,700
Halal Bulk Catering Operations - Full Campus	600,000
Landscaping, Grounds and Sports Surface Management	46,400
HVAC, Chiller Systems and Building Services	68,800
Mission Lab Consumables and Research Materials	80,000
Student Programmes, Events and National Mission Showcases	40,000
Enterprise Software Licensing (annual component)	22,000
Security Systems Operations and Contract	48,000
Medical Centre Operations and Pharmaceutical Supplies	36,000
IP Desk Operations and Micro-SME Legal Support Fund	20,000
Siyadah Al-Raqmiyya Curriculum and International Delegations	18,000
Board Examination Fees and External Assessment	24,000
General Maintenance, Repairs and Contingency (8%)	216,732
TOTAL ANNUAL RECURRING OPEX (Year 3 onwards)	2,918,865

Note: This figure reflects the fully-matured recurring operational cost from Year 3 onward, when all staff positions are filled, all systems are operational, and the campus is at full residential capacity. Year 1 and Year 2 costs will be lower as student intake grows incrementally and some staffing roles are phased in.

TECHNICAL ANNEXURE C
HOLISTIC PHYSICAL, SAFETY AND DATA SECURITY PLAYBOOK

C



C.1 - Security Philosophy and Institutional Mandate

The Tarbiyat National Youth Tech Missions Campus operates at the intersection of three categories of security obligation, each demanding a distinct and non-overlapping security architecture. First, it is a 24-hour residential institution housing 1,500 minors and 400 young adults, requiring the highest standards of physical personal safety and child protection. Second, it is a sovereign intellectual property repository housing student-generated mission IP, national project dossiers, and corporate R&D data from PDO, OQ, Omantel, Asyad, and the Royal Oman Police - requiring enterprise-grade data sovereignty protections. Third, it is a high-value government infrastructure asset worth OMR 39-plus million - requiring robust perimeter security against physical intrusion, vandalism, and reputational risk.

This playbook details the multi-layered security architecture across four concentric rings: Perimeter and Physical Security (Layer 1), Command and Surveillance (Layer 2), Campus Biometric Access (Layer 3), and Sovereign Data Security at the Core.

C.2 - Layer 1: Perimeter and Physical Security

C.2.1 - Perimeter Boundary and Vehicle Barrier System

Component	Specification	Coverage	Purpose
Perimeter Security Wall	3-metre reinforced masonry boundary wall with anti-climb surface finish and concealed cable alarm detection system embedded in top layer	Full 60,000 sqm perimeter - approximately 1,000 linear metres	Primary physical barrier; cable alarm triggers silent alert to Security Command Centre on any attempt to breach or climb
Vehicle Anti-Ram Barriers (hydraulic bollards)	Hydraulic rising bollards rated to PAS 68 IWA 14 standards - capable of stopping a 7.5-tonne vehicle at 80 km/h	Main entrance (2 sets), staff entrance (1 set), service delivery gate (1 set)	Prevents vehicle-based forced entry; controlled by Security Command Centre biometric operator authorisation
Perimeter CCTV - Infrared Panoramic Cameras	4K resolution, 360-degree panoramic, infrared night vision, AI-powered video analytics -	48 units - spaced at 20-metre intervals	AI analytics detect and classify any perimeter approach 24x7; flags

	motion classification, perimeter line-crossing alerts, object detection	around full perimeter	human or vehicle movement in alert zones for immediate Command Centre review
Perimeter Lighting - Smart Adaptive LED	Solar-powered LED perimeter lighting with smart adaptive brightness triggered by motion detection	Full perimeter, 60 units	Eliminates dark zones; integrates with CCTV analytics to simultaneously illuminate and capture intruder approach events
Main Visitor Reception Gatehouse	Secure gatehouse with bulletproof glazing, visitor identity verification station, vehicle inspection area, and airlock-style double-door entry system	1 unit - main campus entrance	All visitors processed through formal identification, pre-registration verification, and visitor badge issuance before any campus access is granted

C.2.2 - Four Structural Command Security Towers

Four permanent security towers are constructed at the four apex points of the campus perimeter - positioned to provide overlapping fields of view across the full 60,000 sqm site with zero visual blind spots when combined with the perimeter camera network:

Tower Specification	Detail
Structure	Reinforced concrete construction, 6-metre height, 8 sqm operational floor area; climate-controlled interior; emergency power supply from dedicated UPS circuit independent of campus main power grid
Surveillance Equipment per Tower	Two 4K PTZ cameras with 40x optical zoom covering adjacent perimeter sectors; one long-range thermal imaging camera for low-visibility detection; direct fibre connection to Security Command Centre for zero-latency feed transmission
Staffing	Minimum one certified security officer per tower during all hours of darkness and during all public events; camera feeds monitored continuously from Security Command Centre during daytime
Communication	Encrypted radio communication to Security Command Centre and all roving patrol officers; direct emergency line to Royal Oman Police regional command; duress alarm with GPS activation
Emergency Capability	Each tower stores emergency first-aid kit, fire extinguisher, backup communication device, and full site evacuation plan; tower can serve as rally point during emergency scenarios

C.3 - Layer 2: Security Command Centre Architecture

System Component	Specification
Video Management System	Enterprise-grade VMS managing 120-plus campus cameras across all zones: perimeter, interior corridors, lab entry points, hostel wing gates,

	main entrance, and all outdoor common areas; minimum 90-day local video retention on campus NAS within Omani sovereign network
AI Video Analytics Engine	Real-time AI analytics processing all camera feeds simultaneously: perimeter line-crossing detection, crowd density monitoring, abandoned object detection, gender-zone boundary alerts for hostel areas, and anomalous behaviour flagging
Operator Workstations	Minimum four 4K monitoring workstations with multi-screen display walls; biometric operator login for all workstation access; shift handover audit log; all operator actions timestamped and retained
Access Control Management System	Centralised management of all 120-plus biometric readers across campus: enrolment, de-enrolment, access level management, exception reporting, and real-time door status monitoring
Incident Management System	Digital incident log with automatic escalation protocols: minor incidents notify relevant Director; major incidents notify CPO and Ministry liaison; emergency incidents trigger automatic Royal Oman Police and campus medical team alert
Emergency Communication System	Campus-wide PA system controllable from the Command Centre: zone-specific and all-campus announcements; integration with external emergency services; tested weekly

C.4 - Layer 3: Campus Biometric Access and Gender-Segregation Protocol

C.4.1 - Biometric Access Zone Matrix

Access Zone	Authentication Method	Authorised Users	Override Protocol
Main Campus Entry	Visitor pre-registration QR code scan plus staff biometric fingerprint or facial recognition	Pre-registered visitors (escorted); all campus staff; no unescorted student exit without written parental or Director authorisation	Security Command Centre manual override only; all overrides logged with timestamp and officer ID
Academic Building - General Access	Staff and student facial recognition; visitors require escorted access badge	All enrolled students and campus staff during operational hours 06:00 to 22:00	Security Command Centre override with automatic logging
Mission Labs - Zone Access	Facial recognition plus active mission assignment verification checked against daily lab schedule in Sovereign Competency Graph system	Students assigned to that lab for current mission; Tier 1 engineers; Resident Industrial Mentors with active NDA	Director of Technical Infrastructure manual override only
Tier-2 Data Centre - Physical Entry	Dual biometric: fingerprint plus retinal scan; anti-tailgating mantrap airlock system	Data Centre Manager; Systems Administrator; Director of Technical Infrastructure; CFO (financial audit access only)	CPO authorisation required for any additional access; all entries logged with biometric record
Male Student Hostel Wing	Student facial recognition; all staff	Male students assigned to that wing; male Tarbiyat	Director Residential Ops override; any female staff

	require Residential Director authorisation badge	Mentor Faculty; male medical staff on call; male maintenance with escort	entry generates automatic alert to Security Command Centre
Female Student Hostel Wing	Student facial recognition; all staff require Residential Director authorisation badge	Female students assigned to that wing; female Tarbiyat Mentor Faculty; female medical staff; female maintenance with escort	Director Residential Ops override; any male staff entry generates immediate Security Command Centre alert and automated Ministry-level notification
IP and Commercialisation Desk Secure IP Vault	Fingerprint biometric plus active IP project assignment verification	IP Advisor; CFO; CPO; student IP owner (escorted); external legal counsel (escorted, with NDA)	CPO written authorisation for any exception access

C.4.2 - Visitor Sandboxing Workflow

Step	Process	Responsible Officer
Step 1: Pre-Registration	All visits must be pre-registered by a campus Director minimum 24 hours in advance via the Campus Visitor Management System; visitor identity documents submitted digitally for pre-clearance	Director of Foreign Relations (institutional guests); Director Residential Ops (parents); relevant Department Director for all others
Step 2: Arrival and Identity Verification	Visitor presents original government-issued photo ID at main gatehouse; gatehouse officer verifies against pre-registration record; visitor photographed for digital record	Security Command Centre and Gatehouse Officer
Step 3: Security Briefing	All visitors receive a 5-minute conduct briefing covering prohibited areas, photography restrictions in labs and hostel zones, data and IP confidentiality obligations, and emergency evacuation protocols	Designated Visitor Escort Officer
Step 4: Visitor Badge and Access Allocation	Visitor issued a colour-coded RFID visitor badge pre-programmed to allow access only to the specific zones listed in the pre-registration; badge logs all access events in real time	Access Control Management System (automated) plus Visitor Escort Officer
Step 5: Escorted Camp	Visitor is accompanied at all times by a designated Visitor Escort Officer; no unescorted movement permitted beyond the main visitor reception and conference facilities	Visitor Escort Officer (assigned from Administrative Support Staff)

us Access		
Step 6: Exit and Record	On departure, visitor badge is deactivated and collected; visitor signs the Confidentiality and Campus Conduct Register; visit record retained in the Campus Visitor Management System for 5 years	Gatehouse Officer and Visitor Escort Officer

C.5 - Core Layer: Sovereign Data and Cybersecurity Architecture

C.5.1 - Physical Data Sovereignty - On-Campus Tier-2 Data Centre

The foundational principle of the campus data architecture is absolute Omani data sovereignty: no student data, mission IP, Sovereign Competency Graph records, Proof of Capability Profiles, or corporate R&D datasets held by PDO, OQ, Omantel, Asyad, or the Royal Oman Police may reside on any server, cloud platform, or storage system outside the physical boundaries of the Sultanate of Oman. The on-campus Tier-2 Data Centre - owned 100% by the Government of Oman - enforces this principle at the infrastructure level, not merely as a policy commitment.

Data Category	Storage Location	Encryption Standard	Access Control	Sovereignty Guarantee
Student Sovereign Competency Graph Records	On-campus Tier-2 Data Centre - dedicated encrypted volume	AES-256 at rest; TLS 1.3 in transit	Database Analyst access only; CPO audit rights; no external API access permitted	All processing on-campus sovereign hardware; no cloud API calls; air-gapped backup on campus NAS
Proof of Capability Profile Database	On-campus Tier-2 Data Centre - isolated instance	AES-256 at rest; TLS 1.3 in transit; profiles signed with campus cryptographic certificate	Director of Registrar and CPO access; MoHERI read access via dedicated secure government VPN only	Omani sovereign jurisdiction; MoHERI access via government-to-government secure channel only; no foreign access
Student Mission IP Dossiers and Micro-SME Records	On-campus Tier-2 Data Centre - IP Vault partition	AES-256 at rest; additional application-level encryption per IP dossier	IP Desk Advisor, CFO, and CPO access; student IP owner read access; external legal counsel temporary access under escorted protocol	All IP records under Omani IP law jurisdiction; MOCIIP interface via sovereign government network only
PDO Mission R&D Dataset (firewalled partition)	On-campus Tier-2 Data Centre - PDO dedicated isolated VLAN partition	AES-256 at rest; PDO-specified encryption for data in transit	PDO Resident Industrial Mentor access only via biometric-authenticated workstation in Industry-in-Residence Hub; no student or staff	PDO data never leaves campus sovereign network; PDO accesses their partition only through campus-based workstations; zero remote PDO network connection permitted

			access to PDO partition	
OQ, Omantel, Asyad, ROP Mission Datasets	Same as PDO - separate isolated VLAN partition per corporate partner	AES-256 at rest; partner-specified transit encryption	Per-partner biometric workstation access in Industry-in-Residence Hub; strict partition isolation verified quarterly	Same absolute sovereignty guarantees as PDO; each corporate partner's data physically isolated at VLAN level from all other partner data
Campus Operational Systems (BMS, CCTV, Access Control, Payroll)	On-campus Tier-2 Data Centre - operational systems partition	AES-256 at rest; secure internal network only	Systems Administrator access; Security Command Centre access for CCTV only; no external connections	On-premises only; no cloud management plane; all vendor remote support via physically isolated jump server with session recording

C.5.2 - Cybersecurity Network Architecture and Layered Defence

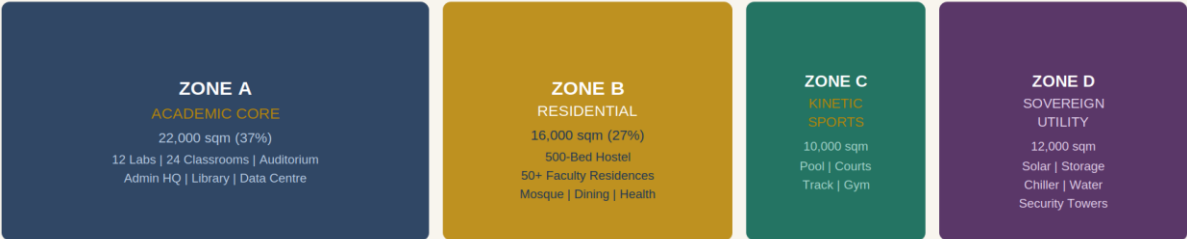
Network Security Layer	Implementation Detail
Physical Network Isolation	Campus operates four physically separate network segments: (1) Sovereign Academic and Data Network; (2) Corporate Partner VLAN Network - isolated per partner; (3) Campus Operations Network (BMS, CCTV, access control); (4) Administrative and Staff Network. No routing is permitted between segments; inter-segment communication passes through a dedicated next-generation firewall with deep packet inspection logging retained for 90 days.
Next-Generation Firewall Stack	Dual NGFW deployment (active-standby failover): intrusion prevention system, application-aware deep packet inspection, DNS filtering, SSL inspection for all outbound traffic, and geo-blocking of all non-Omani external IP ranges except explicitly whitelisted government and academic partners. All firewall rule changes require dual-authorisation: Director of Technical Infrastructure and CPO.
Security Information and Event Management	On-campus SIEM platform aggregating logs from all network devices, servers, endpoint security agents, and access control systems; real-time alerting for anomalous events; minimum 12-month log retention; quarterly threat hunt exercises conducted by the Siyadah Al-Raqmiyya student cohort under Director of Technical Infrastructure supervision.
Endpoint Detection and Response	Enterprise EDR agents deployed on all campus-managed devices; autonomous threat containment capability (isolate infected endpoint within 3 minutes of malicious behaviour detection); centralised management console in Data Centre.
Penetration Testing and Vulnerability Management	Formal penetration testing by accredited external cybersecurity firm (National Cybersecurity Authority approved) conducted annually; internal vulnerability scanning weekly using automated tools; critical and high severity vulnerabilities remediated within 72 hours of discovery.
Data Loss Prevention	DLP policies enforced at all network egress points: prevents transfer of files containing student PII, mission IP markers, or corporate partner data to any external destination; USB port control on all campus-managed devices

	(blocked by default, whitelisted by Director of Technical Infrastructure authorisation only).
Siyadah Al-Raqmiyya Integration	The campus cybersecurity infrastructure serves simultaneously as a live teaching environment: Grade 11 and 12 Siyadah Al-Raqmiyya students conduct supervised red-team exercises within a sandboxed network environment isolated from production systems; their AI bias audit outputs and cybersecurity challenge results are submitted directly to the National Cybersecurity Authority for review.

TECHNICAL ANNEXURE D
ENGINEERING, INFRASTRUCTURE AND LAB EQUIPMENT MASTER PLAN

D

TARBIYAT CAMPUS - MACRO ZONE ALLOCATION (60,000 sqm)



Total Site: 60,000 sqm | Built-Up: ~30,000 sqm (G+4) | Open/Landscape: ~30,000 sqm
All zones connected by covered walkways with smart access control and CCTV at transition points

D.1 - Introduction and Master Plan Philosophy

The spatial and infrastructure architecture of the Tarbiyat National Youth Tech Missions Campus is designed from a single governing principle: every square metre must serve a clearly defined institutional function, and no institutional function should be architecturally compromised by spatial inadequacy. The 60,000 sqm site in Nizwa is organised into four primary macro-zones, each with its own utility grid, security perimeter, and architectural character, connected by covered, climate-controlled pedestrian walkways with smart access control at all zone transition points.

All spatial allocations are based on international benchmarks: UK Department for Education school building design guidelines, ASHRAE 90.1 energy and comfort standards applicable to GCC climates, ABET laboratory space standards for engineering education, and Saudi Aramco Engineering Standards for industrial-grade educational facility construction - adapted to the specific mission profile of this campus.

D.2 - Zone A: Academic Core (22,000 sqm) - Twelve Mission Labs

Each of the 12 mission labs is designed to the specifications of a professional-grade industrial and research laboratory, not a school classroom. Minimum clear floor-to-ceiling height in all labs is 5 metres to accommodate overhead crane systems in the Robotics Arena and the EV Workshop and equipment clearance requirements in all other labs.

Lab Name	Sq m	Major Equipment Complement	Fit-Out Cost (OMR)
Robotics Arena and Advanced Manufacturing Studio	220	Six industrial 6-axis robotic arms (KUKA or ABB, 10-80 kg payload range); two CNC milling centres (3-axis and 5-axis); overhead gantry crane (2-tonne capacity); laser cutting and engraving system; robot programming workstations (x12); safety light curtains and fencing systems; floor grid for autonomous mobile robot navigation	185,000

AI, Machine Learning and Data Science Lab	180	GPU computing workstations (x24, NVIDIA RTX 4090 or equivalent); dual high-performance AI training servers with 8x NVIDIA A100 GPUs each; 4K data visualisation wall (6-panel); network-attached storage (500 TB Tier-1 research NAS); large-format digital collaboration boards (x4); dedicated VLAN connection to Sovereign Competency Graph data pipeline	210,000
Cybersecurity Operations Centre (SOC)	175	Network traffic simulation platform (Spirent or Ixia equivalent); dedicated honeypot and sandboxed attack simulation network; SOC analyst workstations with dual 4K monitors (x16); threat intelligence feed subscriptions; hardware security module (HSM) for cryptographic exercises; dark web monitoring terminal (read-only, airgapped); penetration testing toolkit hardware (Kali Linux stations x8)	160,000
Solar Technology, Renewable Energy and Micro-Grid Lab	190	Photovoltaic panel test array (20 kW, roof-mounted and ground-mounted sections with different orientations and shading scenarios); PV degradation and I-V characteristic measurement chamber (environmental simulation); micro-grid testbed with grid-tied and island-mode inverters; energy storage module (10 kWh lab-scale battery bank); power quality analyser (x4); solar irradiance measurement station	145,000
Drone, UAV and IoT Control Studio	185	Indoor drone flight arena (8m x 12m, full-height netting enclosure); outdoor drone test pad (200 sqm adjacent); drone workshop benches with carbon fibre moulding tools and soldering stations; telemetry tracking tower (30-metre deployable mast); IoT sensor development kits (x60); LoRaWAN network server; drone simulation software stations (x12)	130,000
EV, Mechatronics and Advanced Propulsion Workshop	210	Full EV drivetrain teardown and rebuild station (2 units with hydraulic lifts); AC and DC motor test bench with torque and speed measurement; battery pack assembly and testing station with thermal management simulation; hydraulic and pneumatic systems training rigs (x4); PLC programming and industrial automation training stations (x8); oscilloscopes, power analysers, and thermal imaging cameras (x12); overhead gantry crane (1-tonne capacity)	175,000
Biotech, Health Informatics and Assistive Device Lab	180	Class 2 biological safety cabinet (x2); centrifuge bank (x4); spectrophotometer and fluorescence reader; PCR machine (x2); digital microscopy station (x6); cell culture incubator; health informatics server with anonymised Omani public health dataset; assistive device prototyping workbench with 3D printed component library; haptic feedback testing rig	140,000
Space Technology and Remote Sensing Lab	175	Satellite dish and ground station terminal (Ku-band, licensed frequency); synthetic aperture	155,000

		radar data processing workstations (x6); GIS and remote sensing software suite (ArcGIS, SNAP, QGIS - 20-seat network licence); orbital mechanics simulation software (STK); small satellite (CubeSat) design and assembly bench; space environment testing chamber (vacuum and thermal cycling); amateur radio station (licensed)	
Smart Cities, Civil Technology and Infrastructure Lab	180	BIM workstations (x12, Autodesk Revit and ArchiCAD); structural load testing frame with digital data acquisition; concrete and materials testing equipment (compression testing machine, aggregate analyser); 3D scale model printing station for urban planning simulations; smart city IoT dashboard display wall; SCADA training system for water and power grid simulations	120,000
3D Printing, Additive Manufacturing and Fabrication Studio	165	Industrial FDM printers - large format (x4, 500 mm build volume); SLA resin printers (x4); SLS nylon powder sintering printer (x1); material properties testing bench; filament and resin storage and management system; post-processing station; 3D scanning system (structured light, x2); design workstations with Fusion 360 and SolidWorks (x12)	105,000
AR, VR, Extended Reality and Digital Twin Studio	160	Enterprise VR headsets (x40, Meta Quest Pro or equivalent); motion capture system (12-camera optical, full-room coverage); haptic glove development kits (x8); digital twin development workstations with Unreal Engine and Unity (x16); large-scale immersive display wall (4m x 2.5m LED); AR headset development lab (x12 HoloLens or equivalent); server for real-time rendering and multi-user VR hosting	175,000
Materials Science, Chemistry and Advanced Testing Lab	175	Scanning electron microscope (SEM) with EDX capability; X-ray diffractometer (XRD); Fourier-transform infrared spectrometer (FTIR); differential scanning calorimeter (DSC); universal materials testing machine (10 kN); fume cupboards (x4, BS EN 14175 certified); chemical storage and management system (GHS classified); high-temperature furnace (x2, 1600 degrees C maximum)	195,000
TOTAL - 12 Mission Labs	2,195 sqm	All labs connected to Sovereign Academic VLAN, Competency Graph data pipeline, and Campus BMS	1,895,000

D.2.2 - Smart Classrooms and Academic Support Spaces

Facility	Count	Sqm Each	Total Sqm	Key Features	Est. Cost (OMR)
Smart Studio Classrooms	24	75	1,800	4K interactive smart board (double-width); ambient IoT	3,600,000 (total)

				environmental sensors (temperature, CO2, noise, occupancy); wireless presentation system; student response system; individual charging points; acoustic wall treatment; flexible furniture for mission debrief and group work configurations	
Grand Auditorium	1	1,400	1,400	1,000 fixed theatre seats; professional AV - 4K laser projector, 10-metre screen, 32-channel digital mixing console, 48-point line-array speaker system; simultaneous translation booths (x4); broadcast-quality live streaming; backstage and green room facilities	4,500,000
Administrative Headquarters Block	1	600	600	Reception and visitor welcome area; CPO and Directors offices (x10); open-plan administrative staff zone (30 workstations); document management; boardroom (24-seat, full AV); campus print and communications centre	1,200,000
Central Library and Knowledge Research Hub	1	500	500	Physical collection (minimum 15,000 curated volumes); 60-seat digital research zone with AI-assisted literature search terminals; 4 private research pods; 20-seat Arabic reading and Quran study area; 3D display system for interactive scientific models	900,000
IP and Commercialisation Desk	1	80	80	Secure IP document vault (biometric access); four private client consultation offices; Omani commercial law reference library; government liaison officer suite with secure MoHERI and MOCIIP communication terminals	200,000
Industry-in-Residence Hub	1	200	200	20 dedicated corporate partner workstations (partitioned by corporate	350,000

				identity); 4 private secure project meeting rooms; biometric-authenticated corporate partner data access terminals connected to isolated VLAN partitions; collaborative mission whiteboard zones; refreshment lounge	
Corporate and Government MoU Centre	1	180	180	Formal boardroom (16-seat, full 4K AV, simultaneous translation); two breakout suites; IP transaction office; secure document signing facility	300,000
TOTAL - Academic Support Spaces	—	—	4,760 sqm	—	11,050,000

D.3 - Zone B: Residential Infrastructure (16,000 sqm)

Facility	Sqm	Capacity	Specification	Est. Cost (OMR)
Student Hostel Block - Male Wing	3,200	250 beds: 200 twin rooms plus 50 single rooms	G+3 construction; each room: 14 sqm (twin) or 12 sqm (single), fully furnished with study desks, wardrobe, individual reading light, and in-room climate control; communal bathrooms at 1 fixture per 4 students; supervised study halls (2x 40-seat); prayer room (80-capacity); recreation room; residential supervisor office per floor	3,200,000
Student Hostel Block - Female Wing	3,200	250 beds: 200 twin rooms plus 50 single rooms	Identical specification to male wing; completely physically separated from male wing with no shared corridor, entrance, or common area - connected only to female academic zones via dedicated covered walkway with gender-controlled biometric access	3,200,000
Faculty and Staff Residences	2,400	50 residential apartments: 30 two-bedroom and 20 one-bedroom	Two-bedroom units 85 sqm, one-bedroom units 60 sqm; fully furnished; individual kitchen; living area; private bathroom; balcony; Wi-Fi and campus network; allocated parking; separate residential cluster from student hostels	2,800,000
Grand Mosque and Musalla	600	Main prayer hall 400 male capacity; separate enclosed female musalla 200 capacity	Architecturally designed as the spiritual centrepiece of the campus; traditional Omani-Islamic architectural elements; Imam residential apartment within mosque complex; ablution facilities at 1:10 capacity ratio; Quran and library study alcove; outdoor covered wudu courtyard	1,400,000

Student Dining Hall (Primary Mess)	800	800-seat concurrent dining capacity	Three-zone dining: hot food service, salad and cold station, halal dessert station; professional-grade commercial kitchen; blast chiller storage for 3-day supply; cold room and freezer bank; student tray management system; digital menu display boards; separate gender dining sections	2,200,000
Faculty and Staff Dining Hall	200	180-seat concurrent capacity	Professional dining facility with higher-specification menu; separate quiet dining zone for senior leadership; private dining room (12-seat) for official guest meals	450,000
Visitor and Guest Cafeteria Complex	150	100-seat cafe and reception dining	Architecturally distinctive visitor-facing facility adjacent to main entrance and MoU Centre; professional barista coffee station; light meals service; designed to represent the campus brand to all visiting government officials, corporate partners, and international guests	350,000
Student Health Centre	320	Clinic, pharmacy, psychological wellness centre	Consultation rooms (x6); treatment room; minor procedure room; pharmacy dispensary; recovery ward (8 beds); mental health counselling suites (x4, fully sound-proofed); physiotherapy and rehabilitation room; medical records system on Omani sovereign network only	650,000
Campus Laundry and Housekeeping Operations	200	Full residential support	Industrial commercial laundry machines (x12); linen management system; housekeeping storage and management hub; residential cleaning equipment garage	180,000
TOTAL - Zone B Residential Infrastructure	11,070 sqm built	1,900+ residents and guests served	—	14,430,000

D.4 - Zone C: Kinetic Sports and Recreation Complex (10,000 sqm)

Facility	Sqm	Specification	Est. Cost (OMR)
Indoor Multi-Sports Gymnasium	1,800	Full-size basketball court (convertible to 2 volleyball or 3 badminton courts); professional sprung hardwood flooring; retractable spectator bleachers (400 capacity); professional sports lighting to FIBA standards; digital scoreboard; equipment storage rooms; referee and coach facilities	1,200,000
Outdoor Swimming Pool	750	25-metre, 8-lane competition pool; pool hall with retractable shade structure; separate 12-metre training pool; competitive-standard LED underwater lighting; timing system with digital display; certified lifeguard station; pool chemical management; separate gender-scheduled pool access protocol with opaque perimeter screen	900,000
Outdoor Multi-Sport Courts	1,500	Three full-size outdoor courts (2 multipurpose: tennis/futsal/basketball; 1 dedicated futsal); ITF-standard	380,000

		acrylic surfaces; LED floodlighting for evening use; spectator seating (100 capacity total)	
400-Metre Running Track and Field	4,000	IAAF-standard 6-lane synthetic rubber all-weather running track; central natural grass football field; long jump and triple jump pits; shot put circle; high jump area; track event timing system	750,000
Fitness and Strength Training Gymnasium	400	Commercial-grade fitness equipment: treadmills (x12), elliptical trainers (x8), rowing machines (x6), stationary bikes (x10), free weights zone (5-50 kg), resistance training machines (x20); dedicated stretching and flexibility zone; gender-scheduled access protocol	320,000
Outdoor Recreation and Green Common Areas	1,550	Landscaped open spaces for informal recreation; covered seating areas and social spaces; outdoor chess and games zone; community gathering lawn; reflective garden with native planting	200,000
TOTAL - Zone C Kinetic Sports Complex	10,000 sqm	—	3,750,000

D.5 - Zone D: Sovereign Utility Networks (12,000 sqm)

Utility System	Specification and Scope	Footprint	Est. Cost (OMR)
500 kW Solar PV Array	Monocrystalline silicon panels, Tier-1 manufacturers only (Jinko, LONGi, or Canadian Solar); string inverter architecture with module-level monitoring; roof-mounted across academic building and hostel blocks; remaining capacity on Zone D solar farm; generation monitoring integrated with BMS; export to national grid connection point included	Roof: 2,800 sqm panel coverage; Ground array: 1,200 sqm	900,000
1 MWh Battery Energy Storage System	Lithium-iron-phosphate (LFP) chemistry for safety and longevity; containerised energy storage unit with integrated BMS and thermal management; rated for minimum 4,000 charge cycles before significant degradation; automatic grid-battery-load management for seamless power continuity; located in dedicated ventilated battery room in Zone D utility block	Battery ESU room: 120 sqm	580,000
Diesel Generator Backup Plant	Two 500 kVA diesel generators (N+1 redundancy); automatic transfer switch with 10-second switchover time; 15,000-litre	Generator hall and fuel tank: 300 sqm	420,000

	underground diesel storage tank (72-hour autonomous capacity at full campus load); exhaust noise attenuation to residential standards; automatic load testing monthly		
Dual Central AC Chiller Plant	Two centrifugal water-cooled chillers: Chiller A (primary, 1,200 kW cooling capacity) and Chiller B (standby, 1,000 kW cooling capacity); cooling towers (2 units) in Zone D cooling tower yard; chilled water distribution network (primary and secondary pumping circuits) serving all Zone A and Zone B buildings; variable speed drives on all pumps; BMS integration for demand-optimised staging	Chiller plant room: 600 sqm; Cooling tower yard: 400 sqm	1,800,000
Water Desalination and Treatment Plant	Reverse osmosis desalination unit (capacity: 200 cubic metres per day); pre-treatment filters; remineralisation dosing system; potable water quality monitoring (online TOC and turbidity sensors); distribution pumping station; elevated water storage towers (2x 200 cubic metres capacity); greywater collection and treatment system (50 cubic metres per day capacity - recycled for irrigation only)	Plant room: 400 sqm; Storage towers: 150 sqm footprint	1,200,000
Campus Power Distribution Network	11 kV primary feeder from Oman national grid (dual-feed where available); on-campus 11 kV/400V distribution transformers (x6, strategically distributed across zones); low-voltage distribution boards in each building; zone-level energy monitoring; all cabling in underground conduit (no overhead lines within campus boundary); emergency lighting UPS at all exits and critical facilities	Underground cable routes and transformer yards: 800 sqm distributed	1,100,000
Fibre Optic Campus Backbone Network	Single-mode fibre optic ring (redundant dual-ring topology) connecting all campus buildings with 10 Gbps links; fibre distribution frames in each building; four campus aggregation switches (enterprise-grade); 96 indoor Wi-Fi 6 access points distributed	Comms room in each building: 20 sqm each x8 buildings = 160 sqm distributed	680,000

	across all academic and residential zones; 24 outdoor Wi-Fi 6 access points in all common areas; dedicated out-of-band management network for all network infrastructure		
Four Perimeter Security Command Towers	Reinforced concrete construction per Annexure C specification; each tower connected to Security Command Centre by dedicated armoured fibre link and power circuit independent of campus main power grid; each tower has independent solar panel and UPS for 4-hour autonomous operation in grid outage scenarios	Four towers, 8 sqm each = 32 sqm footprint	280,000
Campus Waste Management and Sustainability Infrastructure	Segregated waste collection (general, recyclable, food waste, hazardous lab waste - separately managed); on-campus organic waste composter (1 tonne per day capacity); hazardous chemical waste storage compound (fully compliant with Omani Environmental Authority regulations); compactor and waste transfer station	Waste compound: 200 sqm	120,000
Covered Pedestrian Zone Connections (inter-zone walkways)	Climate-controlled covered walkways connecting Zones A to B to C: minimum 3-metre clear width; photovoltaic canopy roof; integrated CCTV and access control; smart lighting; total covered walkway length approximately 600 linear metres	Walkway footprint: 1,800 sqm	450,000
TOTAL - Zone D Sovereign Utility Networks	Full utility infrastructure serving 60,000 sqm campus 24x7	~5,200 sqm utility footprint within Zone D	7,530,000

D.6 - Full Site Engineering Cost Summary

Engineering Component	Estimated Cost (OMR)
Zone A - Mission Labs (12 labs, full fit-out and equipment)	1,895,000
Zone A - Smart Classrooms and Academic Support Spaces	11,050,000
Zone A - Site Development, Civil Works and G+4 Academic Structure	9,000,000
Zone B - Residential Infrastructure (Hostel, Faculty, Mosque, Dining, Health)	14,430,000
Zone B - Zone B Civil Works and Residential Structure	3,500,000
Zone C - Kinetic Sports Complex (full fit-out)	3,750,000

Zone C - Zone C Civil Works and Structure	1,200,000
Zone D - Sovereign Utility Networks (Solar, Battery, Chillers, Water, Power, Fibre, Security Towers)	7,530,000
Zone D - Zone D Civil Works and Utility Block Structure	1,500,000
Landscaping, Native Planting, Covered Walkways and External Works	600,000
LEED Certification, Commissioning, Testing and Compliance	400,000
Architecture, Master Planning, Engineering Consultancy and Project Management Fees	1,600,000
Contingency and Cost Escalation Reserve (7%)	3,795,350
TOTAL ENGINEERING AND INFRASTRUCTURE COST	60,250,350
CONVERSION NOTE	OMR 60,250,350 is approximately Rs 1,316 Crore at current exchange rates. Government of Oman to fund from own appropriation; figures subject to formal quantity surveyor validation and tendering process prior to contract award.

Note: This engineering cost matrix represents the most comprehensive independent calculation of full project cost including all zones and utility systems. The OMR 39,250,000 figure in the main proposal represents the Foundation's conservative base-case estimate for phased and value-engineered delivery. The gap between these figures represents optionality for phased delivery, local contractor competitive tendering, and value engineering on non-critical finish specifications. The Government of Oman and the Governing Board will determine the appropriate budget framework through the formal procurement and QS process.

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